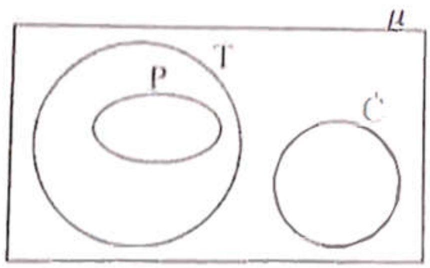
2021 WASSCE CORE MATHEMATICS 2 SOLUTION
Paper Two

1.

$\otimes$	2	3	7	8
2	3	6	5	7
3	6	0	3	6
7	5	3	4	2
8	7	6	2	1

ii) m: m=7

b)i)



ii a) Not valid
B) Valid

2

a)i) P(t + 1)= q -1

Pt + t = q - p

t(p + 1) = q -p

t = $\frac{q-p}{p+\mathbf{1}}$

ii) t = $\dfrac{\frac{3}{4}-\frac{2}{3}}{\frac{2}{3}+1}$

= $\dfrac{1}{20}$

b) 2($\dfrac{2x}{\mathbf{1}-x^2}$) - $\dfrac{2x}{\mathbf{1}+x}$

$\dfrac{4x-2x(1-x)}{(\mathbf{1}-x)(1+x)}$

$\dfrac{2x(1+x)}{(1-x)(\mathbf{1}-x)}$

$\dfrac{2x}{1-x}$

3

a) $\angle MNP = 156/2$

$= 78^{o}$

$m + 204 + 78 + 52 = 360^{o}$

$m = 36 - 334$

$= 26^{o}$

b) Distance $\sqrt{6.5^2 - 3.6^2}$

$= \sqrt{29.29}$

$=5.413$

$=5.41m$

4

(a) Volume of external cylinder = 22/7 - x 8 x 8 x 210

= 42,240 cm ^{3}

Volume of internal cylinder= 22/7 x 7 x 7 x 210

= 32, 340cm^3

Volume of metal used= 42240 - 32,340

= 9,900cm^3

b) |MN| = $\sqrt{(S-0)^2 + (-7-5)^2}$

= $\sqrt{5^2 + (-12)^2}$

= $\sqrt{169}$

= 13 units

5

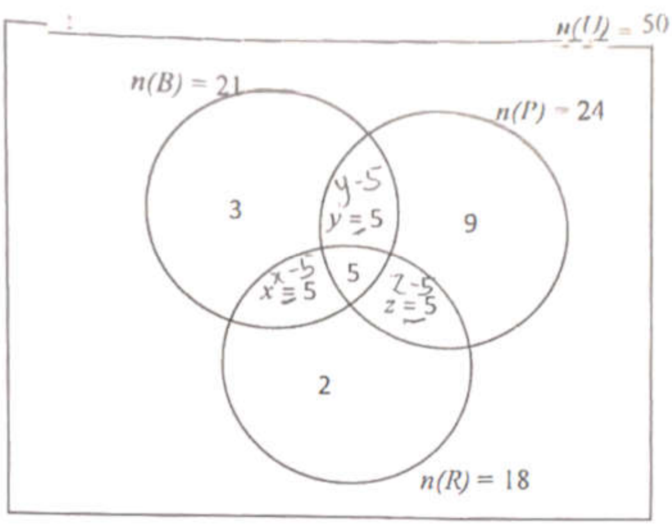
	1	2	3	4	5	6
1	1,1	1,2	1,3	1,4	1,5	1,6
2	2,1	2,2	2,3	2,4	2,5	2,6
3	3,1	3,2	3,3	3,4	3,5	3,6
4	4,1	4,2	4,3	4,4	4,5	4,6
5	5,1	5,2	5,3	5,4	5,5	5,6
6	6,1	6,2	6,3	6,4	6,5	6,6

a) P(odd 1^{st}, 6 on second)
= 3/36
= 1/12

b) P(> 4 on each)
= 4/36
= 1/9

P(total of 9 or 11)
= 4/36 + 2/36
= 6/36
= 1/6 or 0.166

6



a)

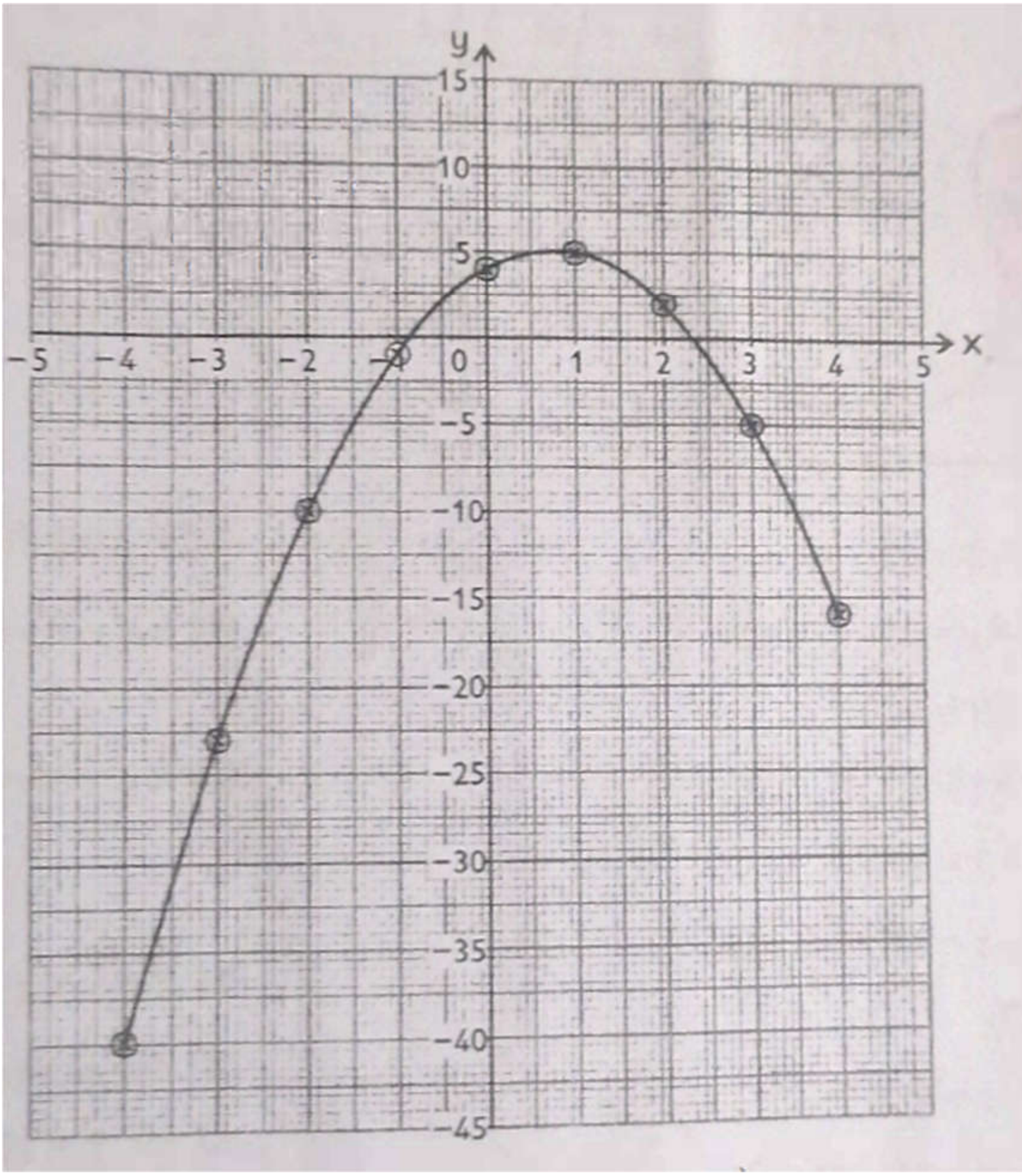
b)
i) (x -5) + (y - 5) + 3 =21
x + y = 23 ------(1)
(y - 5) + 5 (z - 5) + 9 = 24
y + z = 20 -- (2)
(x - 5) + 5 + (z - 5) + 2 = 18
x + z = 21 -----(3)
solving (1), (2) and (3)
X = 12, y = 11, Z = 9
plantain and beans only
= 11 - 5
=6

(ii) exactly 2 kinds = (11 - 5)+(9-5) +(12 - 5)
=17

(iii) None= 50 - (3 + 7 + 5 + 6 + 9 + 4 + 2)
= 14

7

x	-4	-3	-2	-1	0	1	2	3	4
y	-40	-23	-10	-1	4	5	2	-5	-16



b)

c) i) No symmetry
ii) (0.75 ± 0.1)
iii) 0.75 ± ~~0.05~~ < x ≤ 4

8

a) $V_{small} = \pi \times 14 \times 14 \times 600$

$= 117600\,\pi\ cm^3$ or $369{,}000\ cm^3$

$V_{small} = \pi \times 0.25 \times 0.25 \times 14$

$= 9.875\pi\ cm^3$ or $2.57\ cm^3$

NUmber od small cylinder = $\frac{117{,}600\pi}{0.875\pi}$

=134,400

x - tens digit and y - unit digit

x = y+ 1 ----(1)

$\frac{(10x+y)}{x+y} + 8 = 8$

1Ox + y + 8 = 8x + 8 y

2x - 7y = 0 ----(2)

x = 3,y = 2

Number = 32

9

a) Let x represent car

y represent motorcycles

x:y-5:7

y-x-142

x/y = 5/7

x = 5/7y

$\frac{y-7}{7y} = 142 \times 7$

2y = 142 x 7

Y = 497

X = 5/7 X 497

= 355

Cars sold = 355

Saloon cars = 497

b) i) P(only one event) = (3/5 x 7/9) + (2/5 x 2/9)

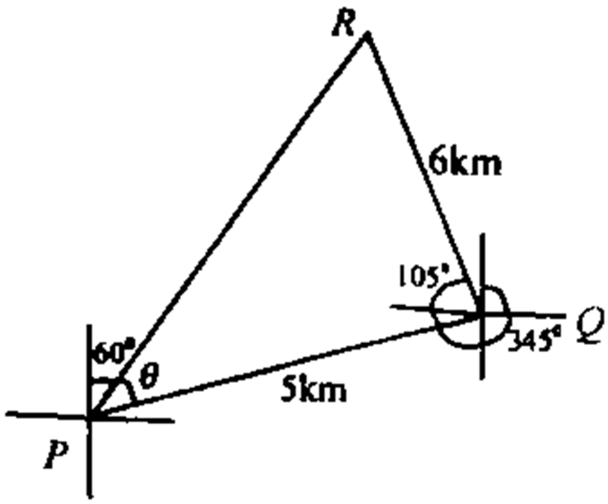
= 21/45 + 4/45

= 5/9

ii P(none of events)= 2/5 x 7/9

=14/45

10



a) i)

ii)

$|PR|^2 = 5^2 + 6^2 - 2(6)(5)\cos 105$

(a) = 61 + 15.529

PR = $\sqrt{76.529}$ = 8.75 km

B) 6/sin θ = 8.75 /sin105

sinθ = 0.6625

θ= 41.49

Bearing = 60 - 41.49

= 18.5°

b) Mean = $\frac{x-2+x+2+4-2x+1+9}{5}$

= $\frac{4x+14}{5} = x + 2$

4x+14 = 5x + 10

= 4x - 5x = 10 -14

x =4

11
a)

Waste	Percentage	Sectoral angle
Plastic	35	35/100 X 360 =126°
Liquid	27.5	27.5/100 x 360 =99°
Metal	17.5	17.5/100 X 360 =63°
Vegetable	12.5	12.5/100 X 360 =45°
Others	7.5	7.5/100 X 360 =27 °
Total	100	360

b) i) LONP = 67°
<MNP = <MOP = 33°
<MNO = <MNP + <PNO
= 33 + 67
= 100°

In △ MPN
<MOP = 33 °
<MPO + 33 + 67 = 180°
<MPO =180 -100 °.
<MPO = 80°.

12

a) △ ORQ is isosceles

i) ⟨RQO =46 °

⟨ ROQ = 180 - (46 + 46)

= 99 °

ii) ⟨ ROQ = 2⟨ RPQ

⟨ RPQ = $\frac{\langle RPQ}{2} = \frac{88}{2}$

= 44 °

(b) Let x be original speed.

Schedule time for 120km = 120/r

After technical challenoe = 120/r-5

Difference in time = 20min = 1/3hr

120/r-5 - 120/r = 1/3

360r - 360(r - 5) = r(r - 5)

r^2 - 5r - 1800 = 0

(r - 45) (r + 40) = 0

Average speed (r) = 45 km/h

13

(a)(i) g(-1) = $\frac{1+3(-1)}{-1-1}$

= 1

(ii) $\frac{1+3x}{X-1}$ =7

1 + 3x = 7x-7

4x = 8

x = 2

(iii) g(-3) = $\frac{1+3(-3)}{-3-1}$

$\frac{-8}{-4}$

= 2

(b)
(i) $a + 4d = a + 7d - 6$
$3d = 6$
$d = 2$

(ii) $5(2a + (10 - 1)2) = 120$
$5(2a + 18) = 120$
$10a + 90 = 120$
$10a = 30$
$a = 3$